ANVESH GUPTA

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SUMMARY

Full-stack developer specializing in the MERN stack (React, Node.js, Express, MongoDB), with hands-on experience designing distributed, algorithmically-optimized backend systems. Competitive programmer (Codeforces Pupil, 400+ DSA problems solved) with applied expertise in graph algorithms and system design.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript

Frontend: React.js, HTML5, CSS3, Responsive Design

Backend: Node.js, Express.js, RESTful APIs, System Design

Databases: MongoDB, Redis

Tools / Technologies: Git, GitHub, Lua, Vercel, Advanced Graph Algorithms (Graph Coloring, Topological Sort)

EXPERIENCE

Teaching Assistant – Data Structures & Algorithms *Jan 2025 – Apr 2025*

- Mentored 80+ undergraduates in DSA as measured by improved assignment accuracy, by resolving code-level doubts, reviewing submissions, and guiding algorithm optimization in C/Java.
- Prepared 80+ students for technical interviews as measured by LeetCode readiness, by structuring targeted problem sets and walkthroughs of core algorithmic patterns.

Freelance Web Developer *2024 – Present*

- Delivered full-stack client web applications as measured by feature completeness and uptime, by building authentication, RESTful APIs, and admin dashboards using React, Node.js, Express.js, and MongoDB.

Club Coordinator – Volleyball Club *May 2025 – Present*

- Organized tournaments spanning 32 teams (192 players) as measured by on-time execution, by planning logistics and scheduling for events including Conquest'25.

PROJECTS

Algorithmic Tournament & Fixture Engine | MongoDB, Express.js, React, Node.js

- Achieved 100% conflict-free round-robin fixtures as measured by zero team double-bookings, by implementing a Graph Coloring algorithm for automated sports scheduling.
- Guaranteed chronological accuracy across dynamic tournament brackets as measured by zero dependency violations, by resolving knockout-stage match constraints with Kahn's Algorithm (Topological Sort).
- Prevented overlapping manual edits as measured by real-time conflict detection accuracy, by building a responsive React dashboard integrated with custom server-side validation via Express REST APIs.

Distributed Rate Limiter Middleware | Node.js, Express.js, Redis, Lua, React, MongoDB

- Optimized memory utilization and burst tolerance as measured by support for 4 rate-limiting strategies, by engineering a framework-agnostic middleware implementing Token Bucket, Sliding Window Counter/Log, and Fixed Window algorithms.
- Eliminated concurrency race conditions in distributed deployments as measured by atomic operation consistency, by integrating a Redis-backed storage layer powered by Lua scripts.
- Visualized real-time API traffic and throttling metrics as measured by live dashboard accuracy, by building a full MERN-stack demo application aggregating MongoDB request logs.

ACHIEVEMENTS

- Ranked in the Top 25 teams nationwide at Hack4Delhi, a national-level hackathon organized by NSUT.
- Solved 400+ Data Structures & Algorithms problems across competitive programming platforms.
- Achieved Pupil rank on Codeforces, reflecting strong algorithmic problem-solving ability.
- Selected participant, Smart India Hackathon (SIH) 2025 — developed "Van Sarthi," an AI-powered forest atlas.

EDUCATION

The LNM Institute of Information Technology — B.Tech, Computer & Communication Engineering *Aug 2023 – May 2027*

CGPA: 7.92 / 10

Shriji Baba Saraswati Vidya Mandir

Class XII: 95.4% | Class X: 95%